Chemistry 3A

Introductory General Chemistry

Experiment 12a

pH of Household Items

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Introduction

- The pH of various household products will be determined in this experiment
- The pH is way of indicating the hydrogen ion concentration: $[H^+]$
- Note that the hydrogen ion $[H^+]$—which is a simple proton, in fact—does not exist by itself, but it is seen as a protonated (ionized) water molecule

$$HA\,(aq) + H_2O\,(l) \rightleftharpoons A^-(aq) + H_3O^+(aq)$$

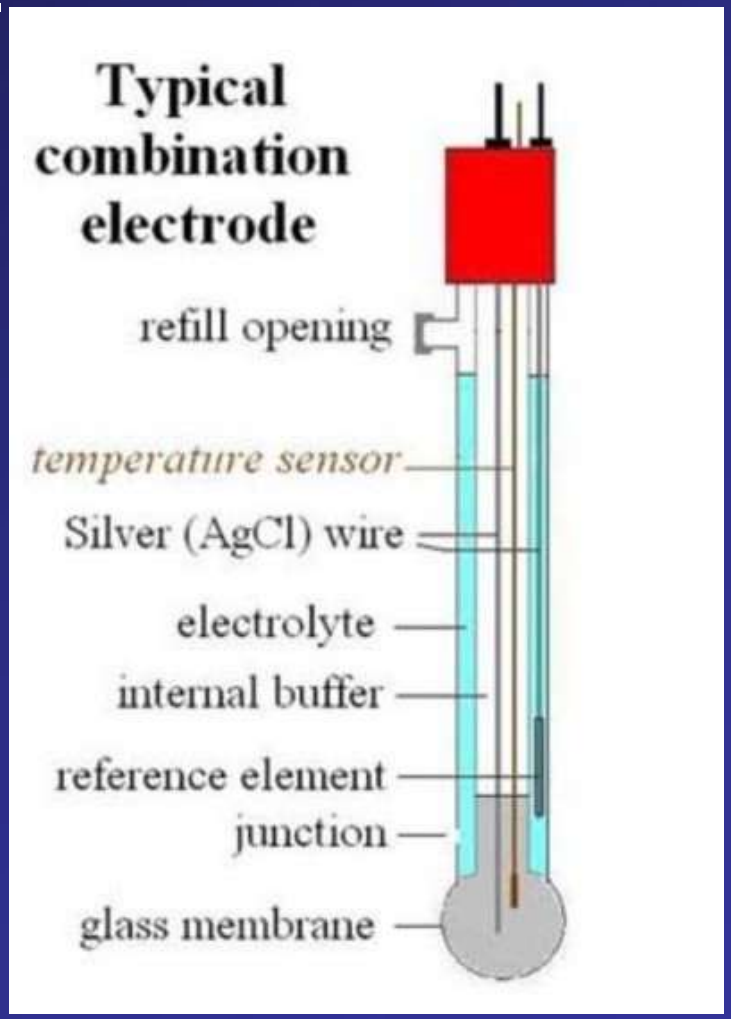
- The generic acid HA has protonated the aqueous solvent molecule H_2O, leaving the conjugate base of HA, A^-, and the conjugate acid of H_2O, the hydronium ion, H_3O^+, as products of the acid-base equilibrium
- pH defined: $pH = -\log [H^+]$
- Get the antilog using math: $10^{-pH} = [H^+]$

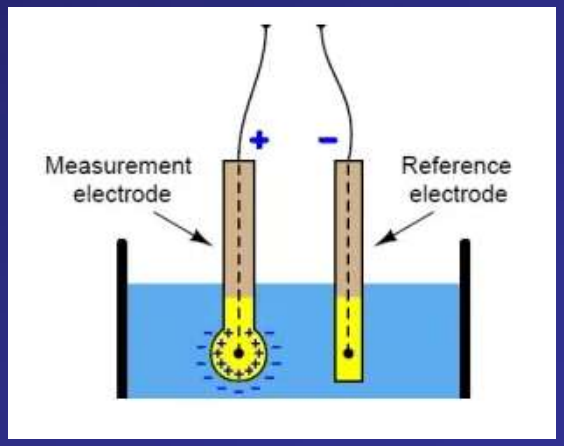
Practice: $[H^+] = 1.0 \times 10^{-7}$ M => $pH = -\log (1.0 \times 10^{-7}) = 7.00$

$pH = 4.92$ => $[H^+] = 10^{-4.92} = 1.2 \times 10^{-5}$ M

Introduction

- The measurement of pH typically in the analytical laboratory is done using $500-1500 of equipment.
- The most important device of the equipment are the electrodes that actually measure the pH: a **pH** or **measurement electrode**, and **reference electrode**
- Nowadays these electrodes are contained in a single unit, the **combination electrode**

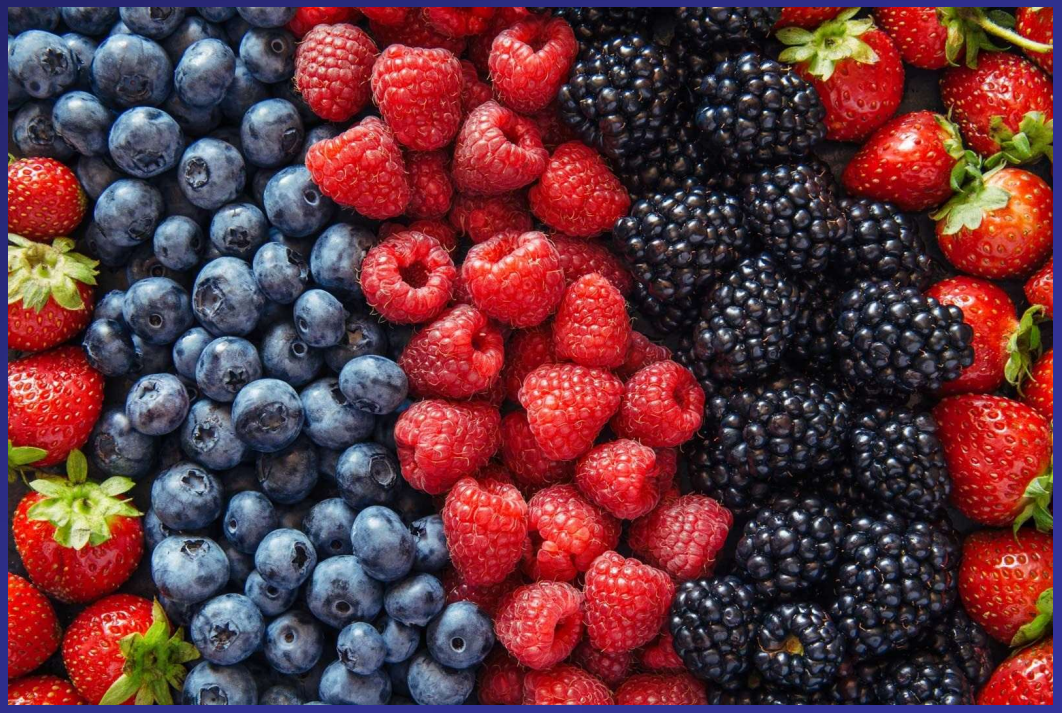




Introduction

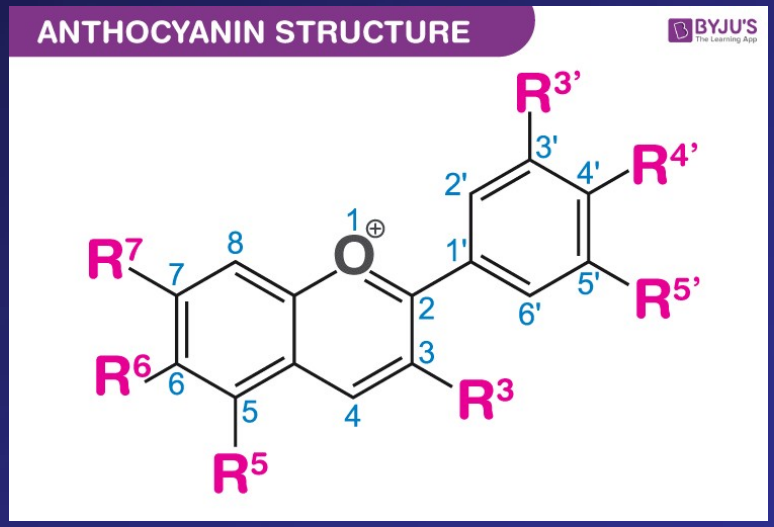
- Other less costly and useful ways of determining pH are **indicator dyes**
- These dyes can be obtained from natural (plant) products which are colorful
- The color can change when the pH is within certain ranges, such as acidic, neutral, or basic

- One class of indicator dyes are **anthocyanins**
- Anthocyanins provide the color to many fruits and vegetables. They in part account for red/crimson leaf color in the autumn

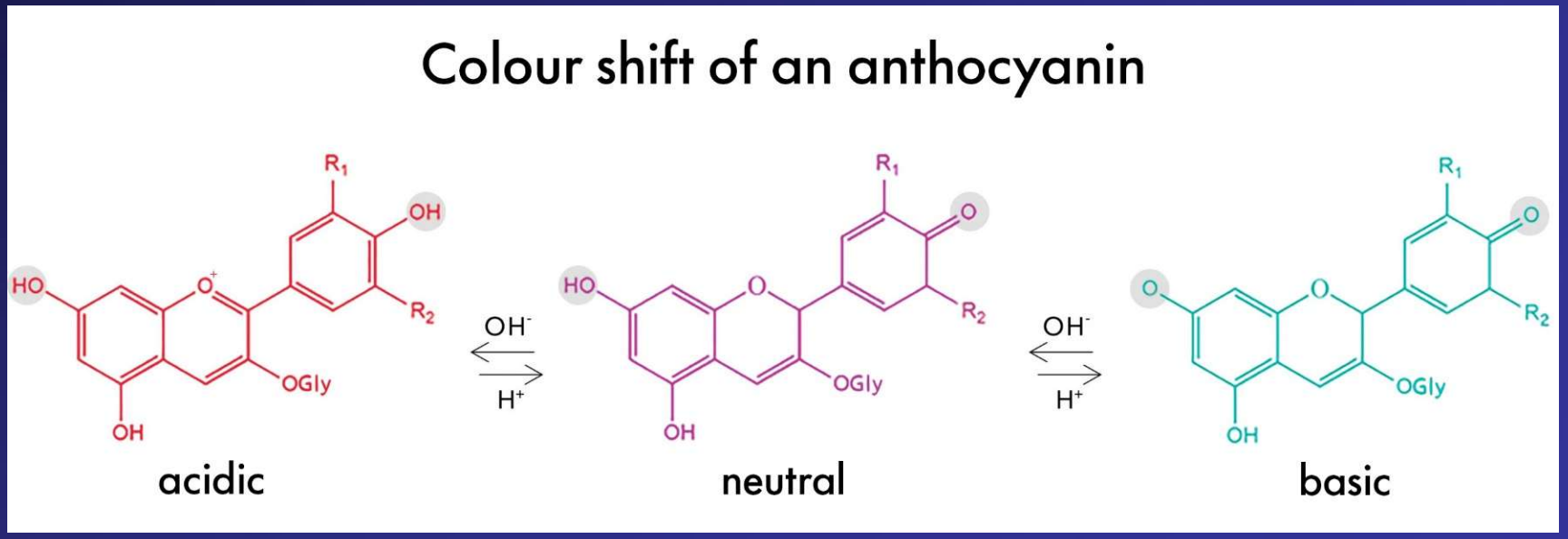


Introduction

- Anthocyanins are polycyclic organic molecules whose structure creates emissions in the colorful visible EM spectrum
- Other classes of biomolecules producing colors include carotenoids and xanthophylls



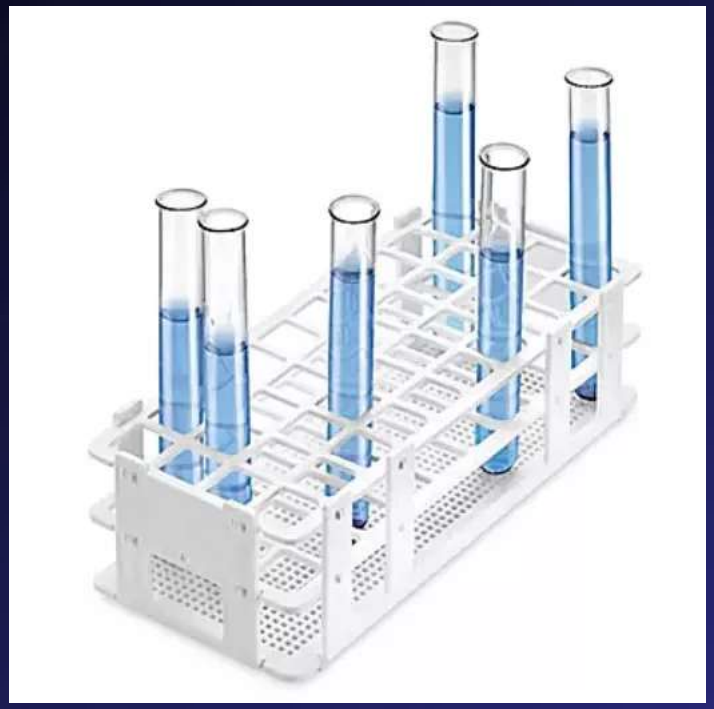
- Below is shown how acidic, neutral, and basic environments create the colors



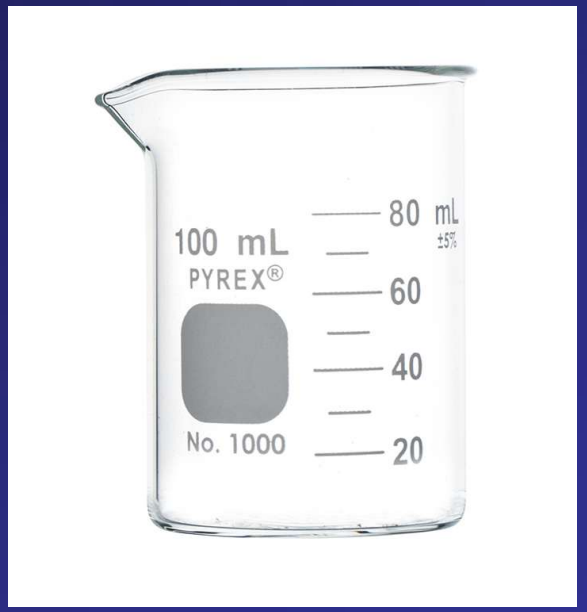
Introduction

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Equipment You Will Use









Consumables You Will Use

- Red cabbage juice
  as a pH indicator dye



Procedure

- Follow procedure in your lab manual

Clean Up

- Discard tube solutions and station solution in inorgance aqueous waste
- Clean tubes with mild soapy water, then rinse well with DI water. Invert tubes on drying rack.

Example Data Analysis

PART 3: CALCULATIONS

Copy the product name and pH from the table on the previous page, then calculate the other information.

*Show your work and keep track of significant figures.

Product Name	$[H_3O^+]$	$[OH^-]$	pH	pOH

Show your work for Station 1 for each of the following:

pH to $[H_3O^+]$:

pH to pOH:

pH to $[OH^-]$:

Example Data Analysis

QUESTIONS

1. For the products tested above,
   a. Which is the most acidic?

   b. Which is the most basic?

POST-LAB QUESTIONS (USE VALUES FROM ONLINE REPORT)

2. Determine the pH of each of the following solutions.
   a. 1.82 g HCl in 0.500 L total solution

      HCl is: A) an acid B) a base

   b. 4.56 g $Sr(OH)_2$ in 0.500 L total solution

      $Sr(OH)_2$ is: A) an acid B) a base

Example Data Analysis

c. 1.82 g HCl and 4.56 g $Sr(OH)_2$ in 0.700 L total solution

1. Write a balanced chemical equation for this reaction:

2. What is the limiting reactant?

3. What mass of the excess reactant is leftover?

4. Find the pH of the solution that results from the excess reactant.

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